

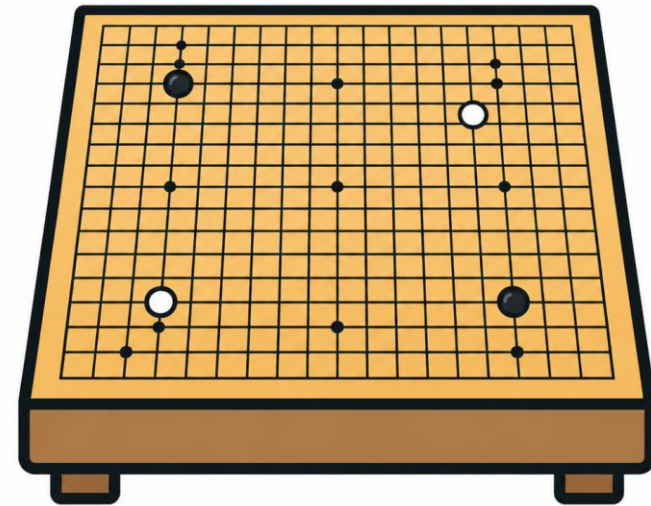
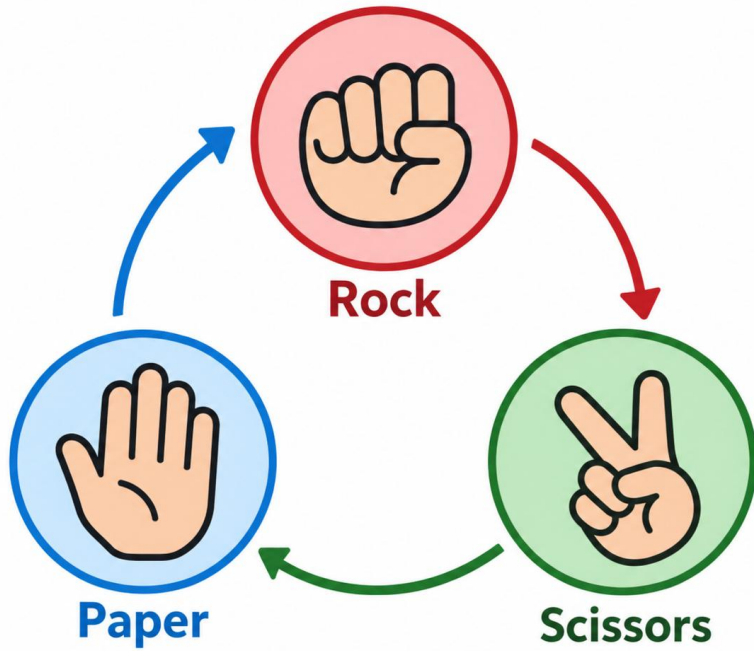
Two-Player Zero-Sum Games

Chen-Yu Wei

References

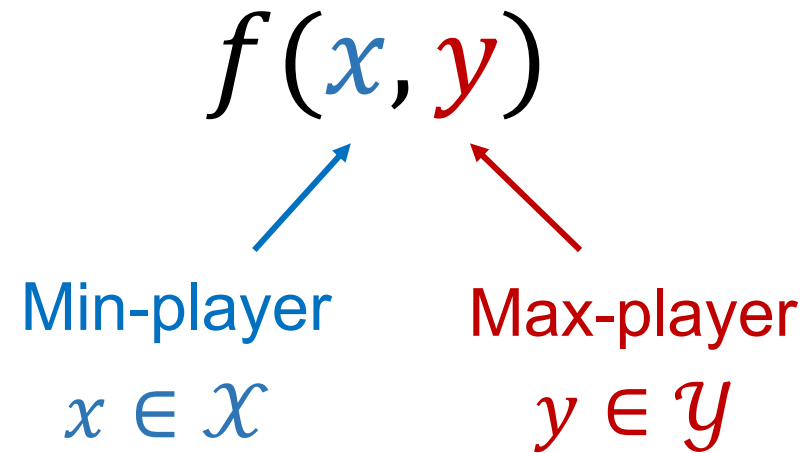
- Lecture 4 of [Haipeng's course](#)
- Lecture 8 [here](#) (for boosting)

Two-Player Zero-Sum Games



Go

Two-Player Zero-Sum Games



$$\mathcal{X} = \{ \text{Rock, paper, scissor} \}$$

$$\mathcal{X} = \Delta_3$$

Game Matrix

$$G = \begin{bmatrix} 1 & 1 & 0 & -3 \\ 2 & 3 & 1 & -5 \end{bmatrix}$$

cost matrix for row-player

Column player

			
	0	0.99 -1	
	-1	0	1
	1	-1	0

$G =$
row player

If row-player uses $p \in \Delta_3$, column player uses $q \in \Delta_3$

$\Rightarrow 0.99$

$\Rightarrow 1$

$\Rightarrow 1$

$$\sum_{i=1}^3 \sum_{j=1}^3 p(i) q(j) G(i,j)$$

$$= p^T G q$$

$$p \in \Delta_N, q \in \Delta_M$$

$$G \in \mathbb{R}^{N \times M}$$

Discussion (Sep 16)

Given a game matrix, (assuming the opponent will do their best), what is the best we can do?

$$\rightarrow \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Compare the Two Values

min x $\max y$ $f(x, y)$

min x $\max y$ $f(x, y)$ $\geq$ $\max y$ $\min x$ $f(x, y)$

$g(x) = \text{my cost if I choose } x$

$h(y)$

depend on f

Protocol: min-player has to tell max-player their decision

min $\hat{x}$ max $\hat{y}$

If we (min-player) have to play under this protocol, then by choosing

$$x^* = \operatorname{argmin}_x g(x) = \operatorname{argmin}_x \left(\max_y f(x, y) \right)$$

my cost is at most

$$\min_x \max_y f(x, y)$$

Proof,

$$\text{my cost} = f(x^*, \hat{y})$$

$$\leq \max_y f(x^*, y)$$

$$= \min_x \max_y f(x, y)$$

max has to tell min their decision

min $\hat{x}$ max $\hat{y}$

their reward $\geq \max_y \min_x f(x, y)$

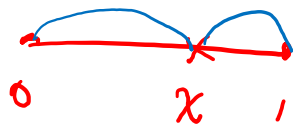
Compare the Two Values

$$\begin{aligned} \text{let } x^* &= \operatorname{argmin}_x \left(\max_y f(x, y) \right) \\ \min_{x \in X} \left(\max_{y \in Y} f(x, y) \right) &= \max_{y \in Y} \left(f(x^*, y) \right) \\ &\geq \max_{y \in Y} \left(\min_{x \in X} f(x, y) \right) \end{aligned}$$

<https://www.youtube.com/watch?v=RmgLPeASjXs>

Minimax Theorem Informal Illustration
(but this is actually not the minimax theorem...)

Compare the Two Values



Game values when both players do their best:

Sometimes this inequality is strict:

① $X = \{\text{rock, paper, scissor}\} = Y$

② $f(x, y) = |x - y|$, $X = Y = [0, 1]$

— | convex in x & y

$$\min_x \max_y f(x, y) \geq \max_y \min_x f(x, y)$$

Protocol: min-player has to reveal their decision (x) to max-player

$$\min_{x \in [0, 1]} \left(\max_{y \in [0, 1]} |x - y| \right) = \min_{x \in [0, 1]} \left(\max \{x, 1 - x\} \right) = 0.5$$

Protocol: max-player has to reveal their decision (y) to min-player

$$\max_{y \in [0, 1]} \left(\min_{x \in [0, 1]} |x - y| \right) = \max_{y \in [0, 1]} (0) = 0$$

Minimax Theorem

$$\boxed{x=y=\Delta_3}$$

$$\sim \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_m$$

(John von Neumann's) Minimax Theorem (1928).

Let $G \in \mathbb{R}^{N \times M}$. Then

$$\min_{p \in \Delta_N} \max_{q \in \Delta_M} p^T G q = \max_{q \in \Delta_M} \min_{p \in \Delta_N} p^T G q$$

$$\min_{x \in X} \max_{y \in Y} f(x, y) = \max_{y \in Y} \min_{x \in X} f(x, y)$$

if X, Y are bounded convex sets, and $f(x, y)$ is convex in x , concave in y

$\forall y, \underline{f(x, y)}$ is a convex func of x

Discussion

Given a game matrix, (assuming the opponent will do their best), what is the best we can do?

$$\begin{aligned} p^* &= \underset{p}{\operatorname{argmin}} \left(\max_{\xi} p^T G \xi \right) \\ &= \underset{p}{\operatorname{argmin}} \left(\max_j (p^T G)_j \right) \end{aligned} \quad \equiv \quad \begin{aligned} &\max_{\xi} \left(\min_p p^T G \xi \right) \\ &\max_{\xi} \left(\min_i (G \xi)_i \right) \end{aligned}$$

Proof by running algorithm

Δ_N : space of distribution over N actions

It suffices to prove

$$\min_{p \in \Delta_N} \max_{g \in \Delta_M} p^T G g \leq \max_{g \in \Delta_M} \min_{p \in \Delta_N} p^T G g$$

- ① Create an expert-learning problem
- ② run no-regret algorithm
- ③ get regret guarantee $\rightarrow$

$$\begin{aligned} p^T G g &= \sum_{i=1}^N \sum_{j=1}^M p(i) g(j) G(i,j) \\ &= \sum_{i=1}^N p(i) \left(\sum_{j=1}^M g(j) G(i,j) \right) \end{aligned}$$

① For $t=1, 2, \dots, T$:

Generate $p_t \in \Delta_N$

Define loss $l_t = G g_t$ ($l_t = p_t^T G g_t$)

② For $t=1, 2, \dots, T$

Generate $g_t \in \Delta_M$

Define reward $r_t = G^T p_t$

$\langle g_t, r_t \rangle = p_t^T G g_t$

Regret for min-player

use exponential weights

$$\sum_{t=1}^T p_t^T l_t - \min_p \sum_{t=1}^T p^T l_t \leq 2\sqrt{T \log N}$$

$\parallel$
 $G g_t$

$$\Rightarrow \sum_{t=1}^T p_t^T G g_t - \min_p \sum_{t=1}^T p^T G g_t \leq 2\sqrt{T \log N}$$

$$\max_g \sum_{t=1}^T \underbrace{g^T r_t}_{G^T p_t} - \sum_{t=1}^T g_t^T r_t \leq 2\sqrt{T \log M}$$

$$\max_g \sum_{t=1}^T p_t^T G g - \sum_{t=1}^T p_t^T G g_t \leq 2\sqrt{T \log M}$$

$$\max_g \sum_{t=1}^T p_t^T G g - \min_p \sum_{t=1}^T p^T G g_t \leq 2\sqrt{T \log N} + 2\sqrt{T \log M}$$

$$\max_{\mathbf{g}} \frac{1}{T} \sum_{t=1}^T \mathbf{p}_t^T \mathbf{G} \mathbf{g} - \min_{\mathbf{p}} \frac{1}{T} \sum_{t=1}^T \mathbf{p}^T \mathbf{G} \mathbf{g}_t \leq \frac{2\sqrt{T \log N} + 2\sqrt{T \log M}}{T}$$

$$\bar{\mathbf{p}} = \frac{1}{T} \sum_{t=1}^T \mathbf{p}_t$$

$$\bar{\mathbf{g}} = \frac{1}{T} \sum_{t=1}^T \mathbf{g}_t$$

Contraction:

If $\min \max - \max \min = V_G > 0$
then let T large enough so that

$$\Rightarrow \max_{\mathbf{g}} \bar{\mathbf{p}}^T \mathbf{G} \mathbf{g} - \min_{\mathbf{p}} \mathbf{p}^T \mathbf{G} \bar{\mathbf{g}} \leq O\left(\sqrt{\frac{\log N}{T}} + \sqrt{\frac{\log M}{T}}\right) \quad \sqrt{\frac{\log M}{T}} \sqrt{\frac{\log N}{T}} < V_G$$

we know $\max_{\mathbf{g}} \bar{\mathbf{p}}^T \mathbf{G} \mathbf{g} \geq \min_{\mathbf{p}} \max_{\mathbf{g}} \mathbf{p}^T \mathbf{G} \mathbf{g}$

$$\min_{\mathbf{p}} \mathbf{p}^T \mathbf{G} \bar{\mathbf{g}} \leq \max_{\mathbf{g}} \min_{\mathbf{p}} \mathbf{p}^T \mathbf{G} \mathbf{g}$$

$$\Rightarrow \min_{\mathbf{p}} \max_{\mathbf{g}} \mathbf{p}^T \mathbf{G} \mathbf{g} - \max_{\mathbf{g}} \min_{\mathbf{p}} \mathbf{p}^T \mathbf{G} \mathbf{g} \leq O\left(\sqrt{\frac{\log N}{T}} + \sqrt{\frac{\log M}{T}}\right) \quad \frac{R_{\text{reg}}}{T} = \frac{o(T)}{T} \rightarrow 0$$

Nash Equilibrium

Def Consider game matrix $G \in \mathbb{R}^{N \times M}$

We call $(p_*, q_*) \in \Delta_N \times \Delta_M$ a Nash Equilibrium iff (ϵ -approximate Nash equilibrium)

$$\begin{cases} \textcircled{1} & p_*^T G q_* \leq p^T G q_* + \epsilon \quad \forall p \in \Delta_N \\ \textcircled{2} & p_*^T G q_* \geq p_*^T G q - \epsilon \quad \forall q \in \Delta_M \end{cases}$$

$$p_*^T G q \leq \max_q p_*^T G q = p_*^T G q_* = \min_p p^T G q_* \leq p^T G q_*$$

Thm. Let $p_* \in \operatorname{argmin}_p \left(\max_q p^T G q \right)$ $q_* \in \operatorname{argmax}_q \min_p p^T G q$

Then (p_*, q_*) is a NE

pf. We want to show $p_*^T G q \leq p_*^T G q_* \leq p^T G q_* \quad \forall p, q$

And know $\max_q \min_p p^T G q \leq \min_p p^T G q_* \leq p_*^T G q_* \leq \max_q p^T G q = \min_p \max_q p^T G q$ By the choice of p_*

Nash Equilibrium

$\bar{p}, \bar{\xi}$ satisfies

$$\max_{\xi} \bar{p}^T G \xi \leq \min_p p^T G \bar{\xi} + \left(\sqrt{\frac{\log N}{T}} + \sqrt{\frac{\log M}{T}} \right) o(T)$$

$$\max_{\xi} \bar{p}^T G \xi$$

$\leq$

$$\min_p p^T G \bar{\xi} + o(T)$$

$$\leq \underline{\underline{\bar{p}^T G \bar{\xi}}} + o(T)$$

$$\bar{p}^T G \bar{\xi}$$

$\Rightarrow$

$$\left\{ \begin{array}{l} \bar{p}^T G \bar{\xi} \geq \max_{\xi} \bar{p}^T G \xi - o(T) \\ \bar{p}^T G \bar{\xi} \leq \min_p p^T G \bar{\xi} + o(T) \end{array} \right.$$

$$\sqrt{\frac{\log(MN)}{T}} \leq \epsilon$$

$$\Rightarrow T \geq \frac{\log MN}{\epsilon^2}$$

Another Proof of the Minimax Theorem

For $t=1, 2, \dots, T$

Choose $p_t \in \arg\min_p p^T G \xi_t$
 $\left(i_t \in \arg\min_i (G \xi_t)_i \right)$

For $t=1, 2, \dots, T$:

① Generate $\xi_t \in \Delta_M$

② Define reward vector
 $\rightarrow r_t = G^T p_t$

③ Update $\xi_t \rightarrow \xi_{t+1}$ using r_t

Useful when

Minimizer has huge # actions
 but, somehow, obtaining

$$\arg\min_p p^T G \xi_t$$

is easy

Regret of max-player: $\max_{\xi} \sum_{t=1}^T p_t^T G \xi - \sum_{t=1}^T p_t^T G \xi_t \leq o(T) = \sqrt{\frac{\log M}{T}} \Rightarrow T \geq \frac{\log M}{\epsilon^2}$

$$\max_{\xi} \sum_{t=1}^T p_t^T G \xi - \min_p \sum_{t=1}^T p^T G \xi_t \leq$$

Follow the same proof as before to show
 $(\bar{p}, \bar{\xi})$ is approx-NE

$$\sum_{t=1}^T \left(\min_p p^T G \xi_t \right) \text{ (by the choice of } p_t) \\ \leq \min_p \sum_{t=1}^T p^T G \xi_t$$

Application: Boosting

Setting: Binary classification

Question: Suppose we can always find a classifier that is only slightly better than random (i.e., 50% accuracy). Can we use this **weak learner** to produce an accurate classifier?

Setting

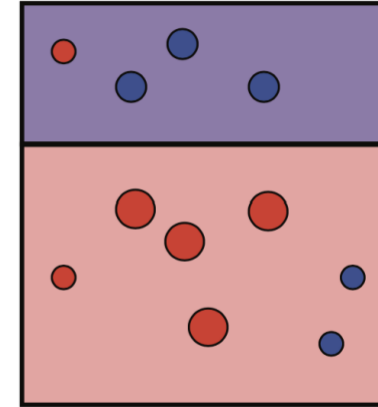
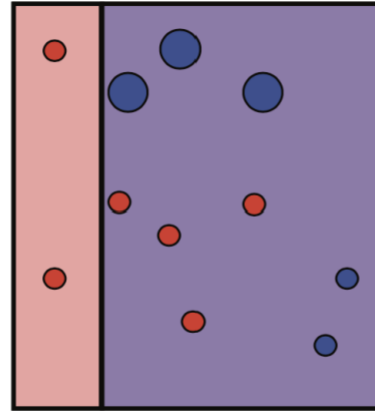
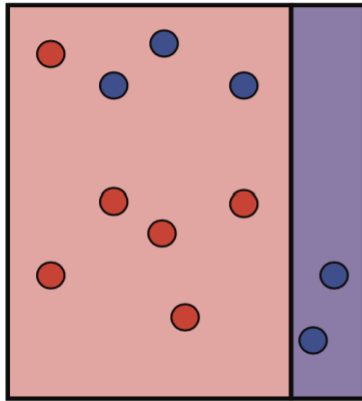
Training data: $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N) \in \mathcal{F} \times \{-1, 1\}$

Classifier: $h: \mathcal{F} \rightarrow \{-1, 1\}$

Loss of a classifier h on a training sample (x, y) :

$$\ell((x, y), h) = \mathbb{I} \{ h(x) \neq y \}$$

What Does “Weak Learner” Mean?



What Does “Weak Learner” Mean?

Assume: $\forall p \in \Delta_N$, $\exists h \in \mathcal{H}$, such that

$$\sum_{i=1}^N p(i) \underbrace{\mathbb{I}(h(x_i) \neq y_i)}_{\ell((x_i, y_i), h)} \leq \underline{\frac{1}{2} - \gamma}$$

Applying Minimax Theorem

$$\frac{1}{2} - \gamma \geq \max_{p \in \Delta_N} \min_{h \in H} \mathbb{E}_{i \sim p} [l(i, h)]$$

$$= \max_{p \in \Delta_N} \min_{\hat{g} \in \Delta_H} \mathbb{E}_{i \sim p} \mathbb{E}_{h \sim \hat{g}} [l(i, h)]$$

$$= \min_{\hat{g} \in \Delta_H} \max_{p \in \Delta_N} \mathbb{E}_{i \sim p} \mathbb{E}_{h \sim \hat{g}} [l(i, h)]$$

$$= \min_{\hat{g} \in \Delta_H} \max_{i \in [N]} \mathbb{E}_{h \sim \hat{g}} [l(i, h)]$$

$$[N] = \{1, 2, \dots, N\}$$

$$\exists \hat{g} \text{ such that } \forall i, \mathbb{E}_{h \sim \hat{g}} [l(i, h)] \leq \frac{1}{2} - \gamma$$

$$\Rightarrow \forall i, l(i, \text{majority-voting}(\hat{g}^*)) \approx 0$$

Applying Minimax Theorem

From No-Regret Learning to Boosting

Input: Training dataset $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$

Initialize a no-regret algorithm **ALG** over Δ_N

For $t = 1, \dots, T$:

Let **ALG** output $p_t \in \Delta_N$

Train a classifier $h_t = \operatorname{argmin}_{h \in \mathcal{H}} \mathbb{E}_{i \sim p_t} [1[h(x_i) \neq y_i]]$

Define **reward** $r_t(i) = 1[h_t(x_i) \neq y_i]$.

Feed reward vector r_t to **ALG**.

Final classifier: $x \rightarrow \hat{y} = \operatorname{sign}(\sum_{t=1}^T h_t(x))$

$$\forall i \in [N] \quad \mathbb{E}_{h \sim g^*} [l(i, h)] \leq \frac{1}{2} - \gamma$$

$$\Rightarrow \text{For all } i, \quad \sum_{h \in H} g^*(h) \mathbb{1}\{h(x_i) \neq y_i\} \leq \frac{1}{2} - \gamma$$

$$\sum_{h \in H} g^*(h) \mathbb{1}\{h(x_i) = y_i\} \geq \frac{1}{2} + \gamma.$$

$\Rightarrow$ aggregate predictor (g^*)

$$x \rightarrow \hat{y} = \begin{cases} +1 & \text{if } \sum_{h \in H} g^*(h) h(x) > 0 \\ -1 & \text{if } \sum_{h \in H} g^*(h) h(x) < 0 \end{cases}$$

↗ +1 or -1

aggregate predictor (g^*) has zero error

Boosting with Exponential Weights

Input: Training dataset $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$

For $t = 1, \dots, T$:

Train a classifier $h_t = \operatorname{argmin}_{h \in \mathcal{H}} \mathbb{E}_{i \sim p_t} [1[h(x_i) \neq y_i]]$

Update $p_{t+1}(i) \propto p_t(i) \exp(\eta 1[h_t(x_i) \neq y_i])$

$$\bar{g} = \frac{1}{T} \sum_{t=1}^T b_t = \frac{1}{T} \sum_{t=1}^T e_{h_t}$$

$$\max_i \mathbb{E}_{h \sim \bar{g}} [\ell(i, h)] \leq \max_i \mathbb{E}_{h \sim g^*} [\ell(i, h)] + \frac{\operatorname{Reg}(T)}{T} \leq \frac{1}{2} - \gamma + \frac{\operatorname{Reg}(T)}{T}$$

To make prediction for new feature:

$$x \rightarrow \hat{y} = \operatorname{majority}(h_1(x), h_2(x), \dots, h_T(x))$$

$$\leq \frac{1}{2} - \frac{\gamma}{2}$$

Choose large T

So that

$$\frac{\operatorname{Reg}(T)}{T} \leq \frac{\gamma}{2}$$

Adaboost

(Yoav Freund and Robert Schapire, 1995)

Input: Training dataset $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$

For $t = 1, \dots, T$:

Train a classifier $h_t = \operatorname{argmin}_{h \in \mathcal{H}} \mathbb{E}_{i \sim p_t} [1[h(x_i) \neq y_i]]$ Let $\epsilon_t = \mathbb{E}_{i \sim p_t} [1[h_t(x_i) \neq y_i]]$

Update $p_{t+1}(i) \propto p_t(i) \exp(\alpha_t 1[h_t(x_i) \neq y_i])$ with $\alpha_t = \ln \frac{1-\epsilon_t}{\epsilon_t}$

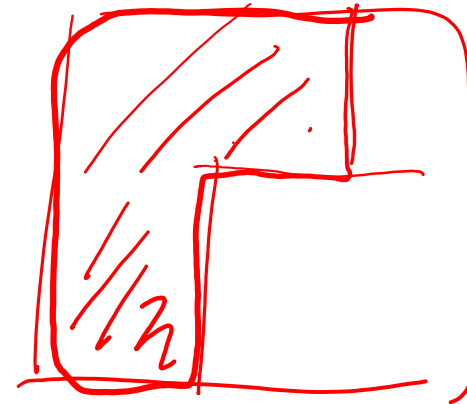
$$\left(\text{Final voting weight for } h_t \right) \propto \alpha_t$$
$$x \rightarrow \hat{y} = \operatorname{sign} \left(\sum_{t=1}^T \alpha_t h_t(x) \right)$$

Example

[Link](#)

Caveat

- We have only focused on training error
- The complexity of the mixture classifier class will increase over rounds – overfitting may happen (and this does happen in practice)
- There are theory arguing that overfitting is controlled (boosting tends to find large-margin classifiers). Lecture 8 [here](#).



Gradient Boosting

Input: Training dataset $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$

For $t = 1, \dots, T$:

Compute (negative gradient $\delta_i = -\partial_2 \ell(y_i, F(x_i))$)

$$\delta_i = y_i - F(x_i)$$

Fit a weak learner $h_t = \operatorname{argmin}_h \sum_{i=1}^N (h(x_i) - \delta_i)^2$

Add small correction $F \leftarrow F + \eta h_t$

Final predictor: $x \rightarrow F(x)$

Summary

- We proved the minimax theorem from no-regret learning. There are two approaches:
 - No-regret-no-regret dynamics
 - No-regret-best-response dynamics
- The proof not only proves **$\min \max = \max \min$** , it ~~always~~^{9/50} provides algorithms to approximate **$\operatorname{argmin} \max$** and **$\operatorname{argmax} \min$**
- Boosting is an application of the No-regret-best-response dynamics



Equilibrium strategy of classifier player